

Avnet Design Services

Safetran Systems

Safetran Systems is a full range supplier of switch machines, railroad wayside signal systems, rail transit signaling and rail-highway grade crossing active warning systems. Products include electro-mechanical signal devices, the electrical/electronic control systems that activate them and communications systems as well as complete signal system design, construction and warehousing.

The company is a worldwide leader in grade crossing technology and warning systems including state of the art constant warning time controls, flashing light and gate warning devices, housings and all associated hardware. In addition to being the market leader in the United States and Canada, Safetran has installations worldwide, including Israel, Mexico, Thailand, Australia, Europe and other countries.

Safetran is the major supplier of vital signal relays to the North American railroads and has a broad range of traditional train control signal devices, power systems, housings and other hardware. Engineered systems are provided to customer specifications and requirements in addition to the component parts.

In addition to signal and train control, advanced telecommunication control systems are offered as well. DTX dispatcher voice communication control is a fully digital system for integrating radio, telephone, intercom and all other voice systems in a single console with quick and easy access to all circuits.

Brian Hogan, is a Product Development Manager at Safetran who works at their electronics division based in Cucamonga, California. This division does research and development, as well as the design and manufacture of electronic equipment for the company. Brian advocates the use of new technology in Safetran products, including the use of FPGAs. He recently was doing some development work with Xilinx. They were looking for some outside design help to meet their end dates. They teamed up with Avnet Design Services (ADS) and explained their requirements.

"The ADS team that was most aligned with our objectives was assigned to our group," Hogan explained. "We had in-house expertise, but just needed some high-level support to make sure we were on the right track. The Xilinx tools provide various predefined components, and there are certain ways of assembling these to optimize for performance in a particular area.

For example, if you are optimizing for speed, you need to do one thing; if you want to optimize for space, you need to do something else. We weren't familiar with it and needed someone to help us through."

Hogan said it really helps to know what you are doing with the Xilinx Project Navigator development tool. For example, the Safetran engineers were having trouble with timing in their design. They had calculated the timing numbers for how fast the signals were moving through the logic, based on the clock rate stipulated in the design. They could then calculate the propagation delays through the various types of logic within the part. From that information they could go in with certain clock rates and see if the design would work.

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*Brian Hogan,
Product Development Manager
Safetran Systems*

"We found we had a disparity. The processing time was in excess of the clock rates we were trying to achieve. There are a couple of ways to reduce the processing times through the Xilinx part. One has to do with knowing which components (e.g. adders, multipliers) have the least propagation delay. The other is choosing which combination or assemblage of these yields the fastest design. This is where we needed someone to just do a reality check on our techniques, and offer criticism and feedback on the approaches we were using. Avnet's regional FPGA Express design team in Texas helped us through the process. A site trial on this particular product was pending and pressure was high; the ADS engineer took a week working with us, but it saved our team an entire month in getting to market. My only complaint is that we didn't take advantage of this earlier in the design cycle so we could have saved even more time. For us, it worked out very well."

Besides saving him money and time, Hogan said the ADS team helped with intangibles too. "This affected how our team was perceived by our customers, both inside the company and out. We had internal and external customers that were waiting for this product, and it really helped that we got it done on time, and with no delays for re-design." Hogan said it really helped his team gain confidence in using this new technology by having the support of the ADS team who had worked with the tool and was familiar with the optimization process. "The level of support you get, it's like graduate level support, like a masters' degree class. It was great."